

FINAL PROJECT PROPOSAL

German Food Safety Monitoring Analytics and Non-Compliance Risk Prediction

Using Official BVL Data

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Programme	Ironhack Data Analytics
Data source	Bundesamt für Verbraucherschutz und Lebensmittelsicherheit (BVL)
Project focus	Food-safety analytics, official monitoring and binary risk prediction

German title

Analyse der amtlichen Lebensmittelüberwachung und Vorhersage des Beanstandungsrisikos anhand offizieller BVL-Daten

SUPERVISOR-READY UPDATED VERSION

1. Project Background

Food-safety authorities conduct official controls to determine whether food products comply with legal, microbiological and chemical safety requirements. Identifying food categories with a higher probability of non-compliance can help authorities and food businesses allocate inspection resources more effectively and take preventive action.

This project will analyse official German food-monitoring data published by the Federal Office of Consumer Protection and Food Safety (Bundesamt für Verbraucherschutz und Lebensmittelsicherheit - BVL). The data originates from official controls conducted by responsible authorities and reported to the BVL. It will be used to investigate patterns in food-safety compliance and develop a machine-learning model that predicts whether a food sample presents an elevated risk of non-compliance.

2. Problem Statement

Official food-control systems generate large amounts of sampling and laboratory data. Systematic analysis is required to identify:

- food categories with high non-compliance rates;
- hazards or substances frequently associated with objections;
- geographical and temporal differences;
- characteristics indicating a higher risk of non-compliance.

A predictive model could support a risk-based approach by identifying samples or product categories that may require greater monitoring attention.

3. Main Research Question

Can information about a food sample - such as its product category, origin, sampling context, region and reporting period - be used to predict its risk of non-compliance?

4. Supporting Research Questions

- 1 Which food and product categories have the highest non-compliance rates?
- 2 Which hazards, substances or microbiological findings are most frequently associated with objections?
- 3 Does non-compliance vary according to product origin?
- 4 Are there significant differences between German regions or reporting years?
- 5 Which available characteristics are the strongest predictors of non-compliance?
- 6 How accurately can machine-learning models identify high-risk food samples?

5. Project Objectives

- 1 Obtain and prepare official German food-control data published by the BVL.
- 2 Examine the quality, structure and completeness of the data.
- 3 Analyse trends and patterns in compliant and non-compliant food samples.
- 4 Identify food categories and control areas with elevated non-compliance rates.
- 5 Create a binary target representing compliance status.
- 6 Develop and compare appropriate classification models.
- 7 Evaluate the models using suitable performance metrics.
- 8 Build an interactive Tableau dashboard.
- 9 Translate the findings into practical food-safety recommendations.

6. Dataset

Primary data source: Bundesamt für Verbraucherschutz und Lebensmittelsicherheit (BVL)

Dataset type: Official German food-monitoring and control data

Depending on the fields available in the downloaded BVL files, the analysis may include:

- food or product category;
- product origin;
- sampling reason or monitoring programme;
- reporting year and responsible region;
- substance, hazard or microorganism examined;
- analytical result;
- official assessment or objection status;
- compliance with applicable requirements.

The exact variables will be confirmed during the initial data-inspection stage.

7. Target Variable

Class	English interpretation	German interpretation
0	Compliant / not objected to	Konform / nicht beanstandet
1	Non-compliant / objected to	Nicht konform / beanstandet

The final target will be constructed from the relevant BVL assessment field after examination of the dataset and documentation.

8. Proposed Data Analysis

8.1 Data preparation

- import the relevant CSV files;
- inspect dimensions, data types, missing values and duplicates;
- translate or decode relevant category codes;
- standardise variable names and combine related tables where necessary;
- remove irrelevant administrative fields and construct the compliance target.

8.2 Exploratory data analysis

- compliant versus non-compliant samples;
- non-compliance rates by product category and origin;
- frequently detected hazards or substances;
- regional differences and yearly trends;
- relationships between sample characteristics and compliance outcomes.

8.3 Data visualisation

Python and Tableau outputs may include bar charts, trend lines, heatmaps, geographical maps, compliance-rate comparisons and interactive filters for year, product, origin and region.

8.4 Machine learning

Classification models may include Logistic Regression, Decision Tree, Random Forest and Gradient Boosting, with an additional model where justified by the data.

Models will be evaluated using accuracy, precision, recall, F1-score, confusion matrices, ROC-AUC where appropriate, and stratified cross-validation. Because missing a genuinely non-compliant sample may have serious consequences, particular attention will be given to recall for the non-compliant class.

9. Prevention of Data Leakage

Columns that directly disclose the final outcome must not be used as predictors. For example, if an analytical-result field already states that a legal limit was exceeded, it cannot be used to predict the official non-compliance decision. The model should use only information reasonably available before the final compliance outcome is known.

10. Expected Deliverables

- 1 A cleaned and documented analytical dataset.
- 2 A Jupyter Notebook containing data preparation and analysis.
- 3 Exploratory data visualisations.
- 4 An interactive Tableau dashboard.
- 5 A machine-learning classification model and performance comparison.
- 6 A confusion matrix with interpretation of false positives and false negatives.
- 7 A presentation of findings.
- 8 A GitHub repository with a professional README.
- 9 Practical recommendations for risk-based food monitoring.

11. Expected Value

The project will demonstrate how data analytics and machine learning can support risk-based food-safety monitoring in Germany. It will help identify food categories, origins or monitoring circumstances associated with higher non-compliance rates.

It also combines the researcher's background in International Food Business and Consumer Studies, food production, quality management for fruits and vegetables, food-safety standards, Python, machine learning, Tableau and data visualisation.

12. Project Summary

This project will use official German food-monitoring data published by the BVL to analyse patterns of food-safety compliance and predict non-compliance risk. Exploratory analysis will identify product categories, origins, hazards, regions and reporting periods associated with higher objection rates.

A binary classification target will distinguish between compliant and non-compliant food samples. Several machine-learning models will be developed and compared using accuracy, precision, recall, F1-score, confusion matrices and stratified cross-validation.

The final output will include a cleaned dataset, Python analysis, a predictive model, Tableau dashboard and practical recommendations. The project aims to demonstrate how data-driven methods can support more targeted and preventive food-safety monitoring in Germany.

Official data source

BVL Open Data - Daten der amtlichen Lebensmittelüberwachung